



## Jacob Stuivenvolt-Allen

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National Center for Atmospheric Research, Boulder, CO

### Research Interests

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Climate variability and change, ENSO dynamics and impacts, climate monitoring and prediction, extreme and impactful weather.

### Education

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**Ph.D. in Climate Science** 2018–2022  
*Utah State University* Logan, UT  
Dissertation: Impact of the Changing Pacific on North American Drought, Atmospheric Rivers, and Explosive Cyclones  
Advisor: Shih-Yu Simon Wang

**B.S. in Environmental Science** 2012–2016  
*Loyola Marymount University* Los Angeles, CA

### Professional Experience

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**Scientist IV** 2025–Present  
*National Center for Atmospheric Research* Boulder, CO

**Postdoctoral Research Associate** 2023–2025  
*Department of Earth and Planetary Sciences, Yale University* New Haven, CT

### Publications

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*Full publication and conference presentation list available on [Google Scholar](#)*

#### In Review

Johnson, Z.F., Chikamoto, Y., Rahimi, S., **Stuivenvolt-Allen, J.**, Mahan, H., Schumacher, B. (2026). Circulation pattern associated with record 2025-26 North American winter may become more common. *Earth's Future*.

**Stuivenvolt-Allen, J.**, Eidhammer, T., McCrary R., Bukovsky, M., Rahimi, S., Chang, H., McGinnis, S. (2026). The North American CORDEX-CMIP6 WRF evaluation run: comparing historical simulations from 25 km to convection-resolving scales. *Geophysical Model Development*.

#### Accepted

15. **Stuivenvolt-Allen, J.**, & Fedorov, A. (2025). Background trade winds and climate sensitivity dictate the pace of warming in the eastern equatorial Pacific. *Science Advances*.

#### Published

14. Chikamoto, Y., Hasan, N., Zhang, W., Meyer, J.J., Gillies, R., **Stuivenvolt-Allen, J.**, McPhaden, M. (2025). Drought Intensification in the Intermountain West During Second-Dip La Niña

Events: Evidence From Utah. *Journal of Applied Meteorology and Climatology*.

13. **Stuivenvolt-Allen, J.**, & Fedorov, A. (2025). Atmospheric nonlinearity controls ENSO asymmetry in a hybrid statistical-dynamical model. *Geophysical Research Letters*. doi:10.1029/2025GL117456

12. **Stuivenvolt-Allen, J.**, Fedorov, A. V., Fu, M., & Heede, U. (2025). Widening of wind stress anomalies amplifies ENSO in a warming climate. *Journal of Climate*. doi:10.1175/JCLI-D-24-0126.1

11. Garuba, O. A., Liu, W., **Stuivenvolt-Allen, J.**, Balaguru, K., Hagos, S., & Wang, H. (2025). The Atlantic Meridional Overturning Circulation's Response to CO<sub>2</sub> Increase: Assessing the Roles of Surface Flux and Oceanic Advection Feedbacks. *Journal of Climate*. doi:10.1175/JCLI-D-24-0143.1

10. Johnson, Z., **Stuivenvolt-Allen, J.**, Mahan, H., Meyer, J., Miksch. (2023). Upper Colorado River streamflow dependencies on summertime synoptic circulations and hydroclimate variability. *Journal of Hydrometeorology*. doi:10.1175/JHM-D-23-0053.1

9. Dalanhesse, L., **Stuivenvolt-Allen, J.**, LaPlante, M., Wang, S.-Y. (2023). An inter-comparison of South American extratropical cyclogenesis using Brazilian Navy data and ERA5. *International Journal of Climatology*. doi:10.1002/joc.8251

8. **Stuivenvolt-Allen, J.**, Wang, S.-Y., Chikamoto, Y., Meyer, J.D., Johnson, Z.F. and Deng, L. (2023). Growing Pacific linkage With Western North Atlantic Explosive Cyclogenesis. *Journal of Climate*. doi:10.1175/JCLI-D-22-0784.1

7. Wright, T., **Stuivenvolt-Allen, J.**, Affram, G., Hasan A., N., Ratterman, C., Zhang, W. (2023). Responses of extreme discharge to changes in surface air and dewpoint temperatures in Utah: Seasonality and Mechanisms. *Water*. doi:10.3390/w15040688

6. Pokharel, B., Sharma, S., **Stuivenvolt-Allen, J.**, Wang, S.-Y., LaPlante, M., Gillies, R. R., Khanal, S., Wehner, M. F., Rhoades, A. M., Hamal, K., Liu, W.-Y., Mukherjee, S., & Aryal, D. (2023). Amplified drought trends in Nepal increase the potential for Himalayan wildfires. *Climatic Change*. doi:10.1007/s10584-023-03495-3

5. **Stuivenvolt-Allen, J.**, Wang, S.-Y. (2022) North American fire weather catalyzed by the extratropical transition of tropical cyclones. *Climate Dynamics*. doi:10.1007/s00382-022-06561-1

4. **Stuivenvolt-Allen, J.**, Wang, S.-Y. S., Johnson, Z., & Chikamoto, Y. (2021). Atmospheric rivers impacting Northern California exhibit a quasi-decadal frequency. *Journal of Geophysical Research: Atmospheres*, 126, e2020JD034196. doi:10.1029/2020JD034196

3. Morgan, B., Spangler, K., **Stuivenvolt-Allen, J.**, Morrisett, C. N., Brunson, M. W., Wang, S.-Y. S., & Huntly, N. (2021). Water Availability for Cannabis in Northern California: Intersections of Climate, Policy, and Public Discourse. *Water*, 13(1). doi:10.3390/w13010005

2. **Stuivenvolt-Allen, J.**, Wang, S.-Y. S., LaPlante, M. D., & Yoon, J.-H. (2021). Three Western Pacific Typhoons Strengthened Fire Weather in the Recent Northwest U.S. Conflagration. *Geophysical Research Letters*, 48(3), e2020GL091430. doi:10.1029/2020GL091430

1. **Stuivenvolt-Allen, J.**, & Wang, S.-Y. S. (2019). Data-mining climate variability as an indicator of U.S. natural gas. *Frontiers in Big Data: Integrating Big Data with Earth System Models of Natural and Human Systems*. doi:10.3389/fdata.2019.00020

## Selected Presentations

**[Invited] Stuivenvolt-Allen, J.**, Eidhammer, T., McCrary R., Bukovsky, M., Rahimi, S., Chang, H., McGinnis, S. (2026). "How Far Can 12 km Take Us? Comparing historical simulations from 25 km to convection-resolving scales." American Geophysical Union, New Orleans.

**[Invited] Stuivenvolt-Allen, J.**, & Fedorov, A.V. (2025). "Atmospheric Nonlinearity Controls ENSO Asymmetry in a Hybrid Statistical-Dynamical Model" Princeton GFDL Seminar.

**Stuivenvolt-Allen, J.** (2025). "Impacts of the North American Circulation Response to Western North Pacific Tropical Cyclones" Seminar, Research Applications Laboratory at NSF NCAR.

## Active Proposals

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- **NCAR PI:** Exploring the collapse and recovery from abrupt shifts in the West African Monsoon. Under review by the United Kingdom Advanced Research and Innovation Agency.
- **NCAR PI:** An Integrated Research Center for Energy Structure Resilience to Weather Hazards. Under review by the Department of Energy.
- **PI:** Constraining the drivers of ENSO variability and asymmetry with NASA observations. Under review by NASA.
- **PI:** Trajectories of the 2020-2023 Southwest US Drought: Integrating E3SM ensemble simulations with a water planning and demand model. Under review by the Department of Energy, Earth-Energy Systems Modeling subgroup.
- **Co-PI:** Evaluating the utility of mid-century climate projections for infrastructure resilience. **Selected for funding by the NOAA Climate Projections Office.**

## Awards & Honors

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**2022:** Doctoral Researcher of the Year: Plants, Soils, and Climate Department, Utah State University  
**2021:** Top Award for Science Writing from the American Society of Journalists and Authors: The Butterfly and the Blaze  
**2020:** Graduate Enhancement Award, Utah State University  
**2019:** Best Research Presentation: Plant, Soils and Climate, Utah State University  
**2017:** Utah Master Naturalist: Wetlands Certification  
**2017:** Utah Master Naturalist: Mountains Certification  
**2016:** Sigma Xi National Research Honor Society Member

## Professional Activities

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### Service

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- Peer reviewer for GRL, ERL, JCLIM, CLIM-DYN, JGR-A, JAMC, etc.
- Mentor for NSF Significant Opportunities in Atmospheric Research and Science.
- Seminar organizer for Yale Department of Earth and Planetary Sciences (2022-2024).

### Outreach and Communication

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- Volunteer weather forecaster for Utah Public Radio (2018-2022)
- Scientific journalism:
  - Improving Our Relationship With Fire - An ARCGIS Storymap
  - The Butterfly and the Blaze - Eugene Weekly
  - Utah Water Issues - A Blog on Utah's Water Supply and Management
- Website development and climate monitoring: <https://jacobstuallen.github.io>

## Technical Skills

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**Programming Languages:** Python, R, MATLAB, IDL, NCL, Bash/Shell scripting, JavaScript

**Global and Regional Models:** CESM, E3SM, WRF, MPAS-A

**Scientific Computing:** Parallelized workflows, HPC environments, job schedulers (SLURM, PBS)

**Development Tools:** Git/GitHub, LaTeX, HTML/CSS, version control

## References

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Available upon request